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Objective :-

To Study & running Automatic Injection Moulding Machine.

Machine name :- Microprocessor Controlled Automatic Injection Moulding Machine.

Material name :- PPCP

Tools use :- Silicon Spray, Brush and Spanner Raw Material for Product Making

Principle :- To other words it melts the plastic raw material inject the Molten plastic in mould and from the disined product.

Machine Specification :-

Clamping Force - 100 Tones
Make :- S S T Engines Pvt LTD
Shot Capacity - 180 gram
Clamp Stroke - 360 MM
Screw G/O Speed :- 198 RPM

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Screw C/O Ratio — 20:01
Injection Speed — 28 mm/s
Screw SIZE — 38 MM

Parts and Fancation:

Hopper — It is used to store the Raw material

(1) Hopper dryer — it is used to store proceed the Material.

(2) Reciprocating Ra Screw — it is used to Mix the Material in a proper manner. after that it inject the Material in the Mould and fill the Material for next shot.

(3) Band Heater — it is used to generate the heat for melting the material

(4) Barrel — it Just like a hollow Cylinder which helps up to resides the Melted Material

(5) NOZZLE — it is used for inject the Melted Material in the Mould with high processor

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(6) NRV :- NRV is used for ~~processing~~ preventing back flow of melted material during injection ~~Processure~~ time.

(7) Hydro Moter :- it is used for rotating the reciprocating screw.

(8) Hydrolic Cylinder :- it is used for backward and forward injection unit.

(9) Servo Moter → it is not ~~Coorse~~ ~~Run work~~ Continuously when it is needed it will run.

(10) Mani fold :- it is used for distribution of hydrolic oil and reverse hydrolic oil in tank.

(11) Heat Exchanger :- it is used for cooling the hydrolic oil.

(12) FRL :- filter regulation and lubrication compresure air system are used to deliver clear air ~~also~~ fixed pressure and it also increase their opration lifetime.

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(13) LVDT– Linear variable differential transfer is an electro mechanical sensor use to measure the position.

(14) Locating Ring :- it is used to locate center nozzle and mould ~~Sprue~~ ~~Bush~~ Sprue Bush.

(15) Mould :- it is used give shape and size of the product.

(16) Stationary Plate– it is used for opening and closing the mould and also for fixing the core volve on it.

(17) Tie bar → it is used to movable plate on it and connect the stationary platen moving platen and tail platen.

Procedure :-

(1) Clean the machine properly with cotton clotes.

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(2) Set the temprature at the heater manually according to material requirments or specification and wait for material melt temprature.

(3) After heating the barrel purging out the waste material or material which feeded byor it.

(4) Inject the melted material into the mould through nozzle and held on 2-3 seconds.

(5) Switch on the machine.

Trouble & Shoting –

	Defect	Causes	Remendies
(1)	Shot – Shot	Injection pressure injection speed is low.	Increase injection pressure increase injection speed.
(2)	Flash	Injection pressure to high injection speed is high	Decrease injection pressure. Decrease the injection speed.

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Defect	Causes	Remendies
③ Shrinkage	Injection pressure is low. Melt temprecture to high	Increase the Injection pressure. Decrease in Melt temprecture
④ Burn marks	Injection speed to high. ~~Injection~~ proper gate location.	Decrease Injection speed. Change the gate location
⑤ Weld lines	Injection speed to low. Mould temp. to low.	Increase the Injection speed. Decrease temp

Safety.

① Do not touch to the barrel.
② Don't Clens in the Machine when running
③ Always Wear Safety goggles glaves and leather shoes
④ Always Carry first aid-kit.
⑤ Don't temprecture with the Machine Safty devices.

Conclusion:→

We have successfully studies about automatic Injection moulding machine and it part function.

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Injection Machine Started

1. सबसे पहले MCB को ON करना है।
2. मशीन का power Batan on करना है।
3. Heater on करना है।
4. Motor on करना है।
5. Mould open करना है, अगर mould बंद रहे तो
6. Nozzle forword करना है। / back
7. Material Check करना है।
8. → Semi Auto में machine को करना है।
9. → barrel से Material को purging करना।
10. Injection unit - Nozzle को आगे ले जाना।
11. Mould को बंद करना।
12. Injection करना।
13. Cooling करना।
14. Product के अनुसार Auto या Semi Auto में मशीन को चलाना।

मनोवैज्ञानिक भावनात्मक रूप से व्यक्ति यह पीड़ित है
व्यक्ती

संवेगात्मक

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